

Marco Israel Rodríguez Cornejo

📍 Ixtapaluca, México ✉ marco_israel@ciencias.unam.mx ☎ 52 55 8150 14 58 🌐 marcoisrael.github.io
in marco-israel-rodriguez-cornejo-b6949a276 🐙 marcoisrael

Profile

I am a physicist focused on data science, numerical simulations, and machine learning. During my training, I conducted simulations of complex physical phenomena using various programming languages and tools such as Python and Fortran at the Institute of Nuclear Sciences' high-performance data center.

Education

Facultad de Ciencias UNAM <i>Bachelor's degree in physics</i>	<i>2017 - 2022</i>
Instituto de Ciencias Nucleares UNAM <i>Intern</i>	<i>2022 - 2023</i>
Tecnológico Nacional de México <i>Data science certificate</i>	<i>2023 - 2024</i>

Experience

Latin American Journal of Science Education Amplitudes grandes de un péndulo simple amortiguado 🔗	<i>August 2017</i>
I collaborated on an article on numerical simulations in a physical system for the Latin American Journal of Science Education with the advice of Dr. Alejandro González y Hernández. I conducted a controlled experiment in the laboratory and wrote a program to solve the problem using the second-order Euler method.	
Instituto de Ciencias Nucleares UNAM Social work	<i>September 2022 - June 2023</i>
I did social work at the Instituto de Ciencias Nucleares ICN. My work was perform a set of numeric simaltions in the ICN's super computer, for the project "Materia fuertemente acoplada en condiciones extremas".	
Instituto de Ciencias Nucleares UNAM Enfriamientos rápidos del modelo 2d O(3)	<i>March 2023 - July 2025</i>
My thesis project is a study of rapid cooling in a physical model called 2d O(3). The research required the application of Monte Carlo methods, Markov chains, and advanced statistics, as well as the development of an efficient program capable of performing the simulations, guided by Dr. Wolfgang Peter Bietenholz.	

Skills

Python (90%): numpy, pandas, scikit-learn, tensorflow, pyspark, matplotlib, regex, urllib, Pillow.
Fortran (90%): gfortran.
Bash Shell (80%): Git, SSH, FTP, Docker, qemu, ffmpeg.
SQL (80%): Mysql, Mariadb, SQL server.
Java (70%): OpenJDK.
Software adicional: Ms office, VS code, Power Bi, Vim, JupyterLab, Spyder, Latex, Linux, Windows.
Idiomas: Español, Inglés(80%)

Statistical methods. I have worked with statistical models such as $O(n)$, Ising, and van der Waals. At their core, these models can be applied to a wide range of systems: economic growth, transport dynamics, population growth, virus propagation, and more.

Numerical methods. Numerical methods are the way to solve dynamic equations and answer the questions of the past, the present, and the future of a system. I have wide experience with the application of numerical methods in order to solve differential equations, numerical integration, Fourier transforms, and linear regression.

I can implement Monte Carlo methods as a solution for common problems in areas such as economy or geography, where there are multidimensional systems and millions of variables.

Data extraction. Information extraction from sources such as: plain text files, HTML, JSON, CSV, XML, SQL databases, audio, video, image, and other non-structured data formats.

Data structure. I have experience with big databases and information extraction in high-performance systems. I can use statistical models such as linear regression, decision trees, clustering, and neural networks to extract relevant information or test the performance of the models.

Values. I have experience working with diverse teams. I learned to communicate myself in a positive and precise way, working collaboratively and adapting to unexpected situations. My ethics are based on honesty, responsibility, great curiosity, and critical thinking.